



The Agentic Paradigm Shift

The Agentic **Paradigm Shift**

Agents don't finish your sentences - they finish your goals by planning steps, using tools, and verifying results.



Why it Matters

- Plan → Act → Verify
- Tools to prove changes work
- Human-in-the-loop



The Old Model

- Token-by-token suggestions
- One-off snippets, no execution
- You verify manually
- Easy drift & rework

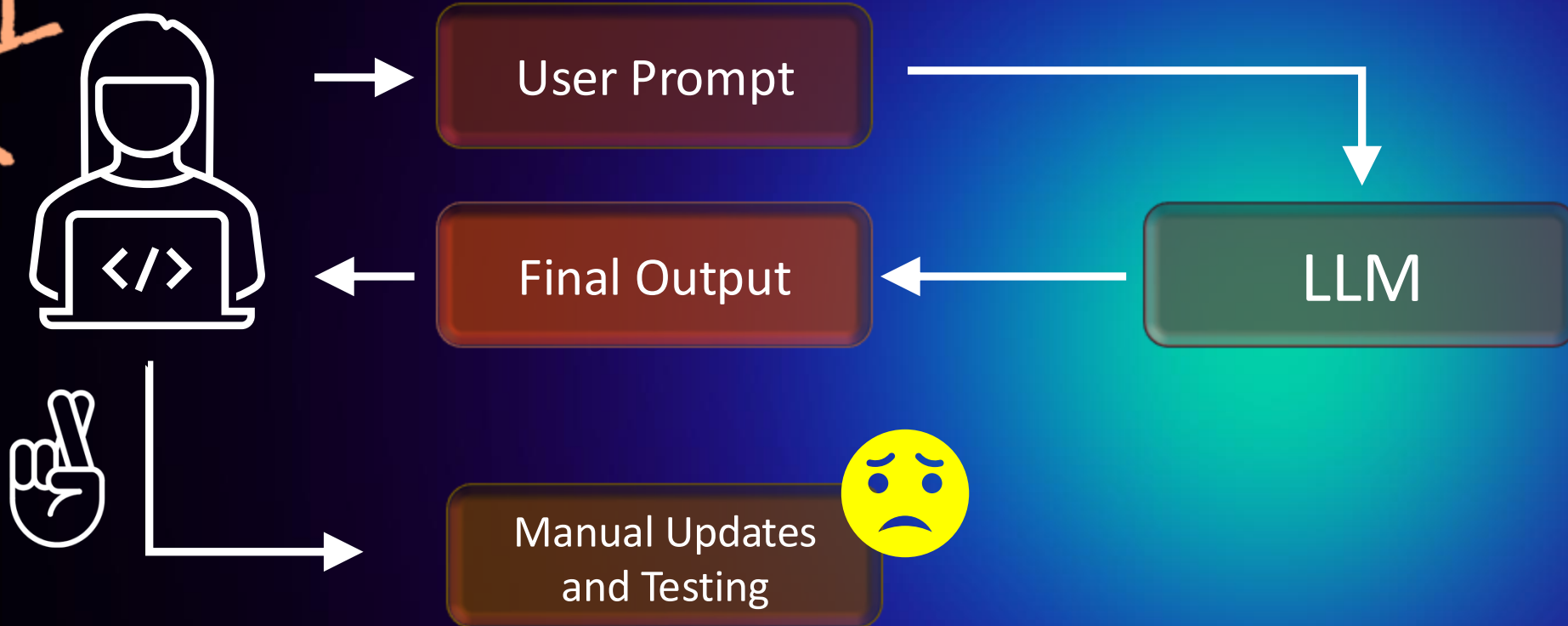


Agentic Model (Coding Agent / Agent Mode)

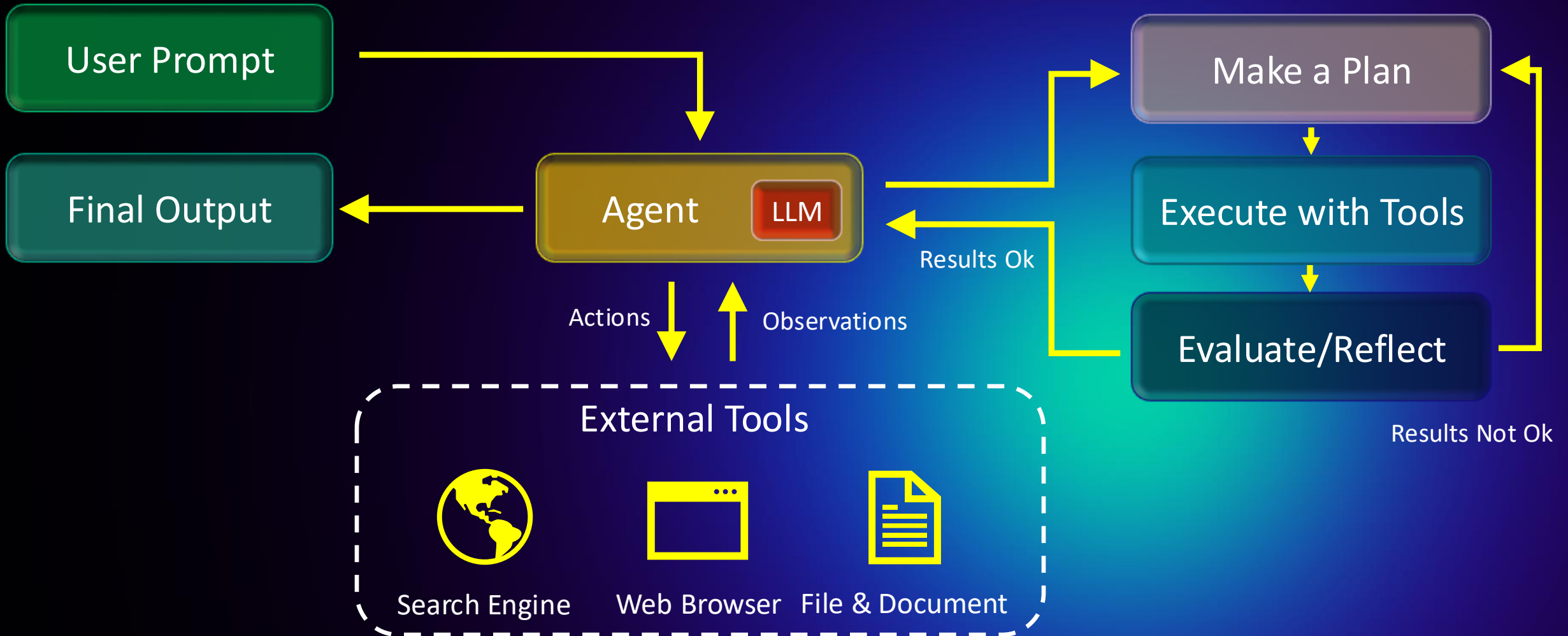
- Goal → plan → multi-step run
- Calls tools (test, build, refactor)
- Diffs + checkpoints for review
- Repeatable tasks, less toil

Pre-Agentic Workflow: **Prompt-In, Output-Out**

START
OVER



The Agentic AI Agent Workflow



What is an Agent?

AN LLM-powered controller that plans, calls tools, observes results, and replans until done.



The Controller

- Plans steps, select tools, executes
- Uses context & memory to iterate and improve
- Evaluates results - replans until done



Definition of Done

- Objective checks pass (tests, linters, build)
- Reproducible trace (plan, commands, diffs)
- Acceptance criteria met and summary generated

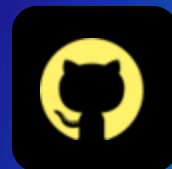
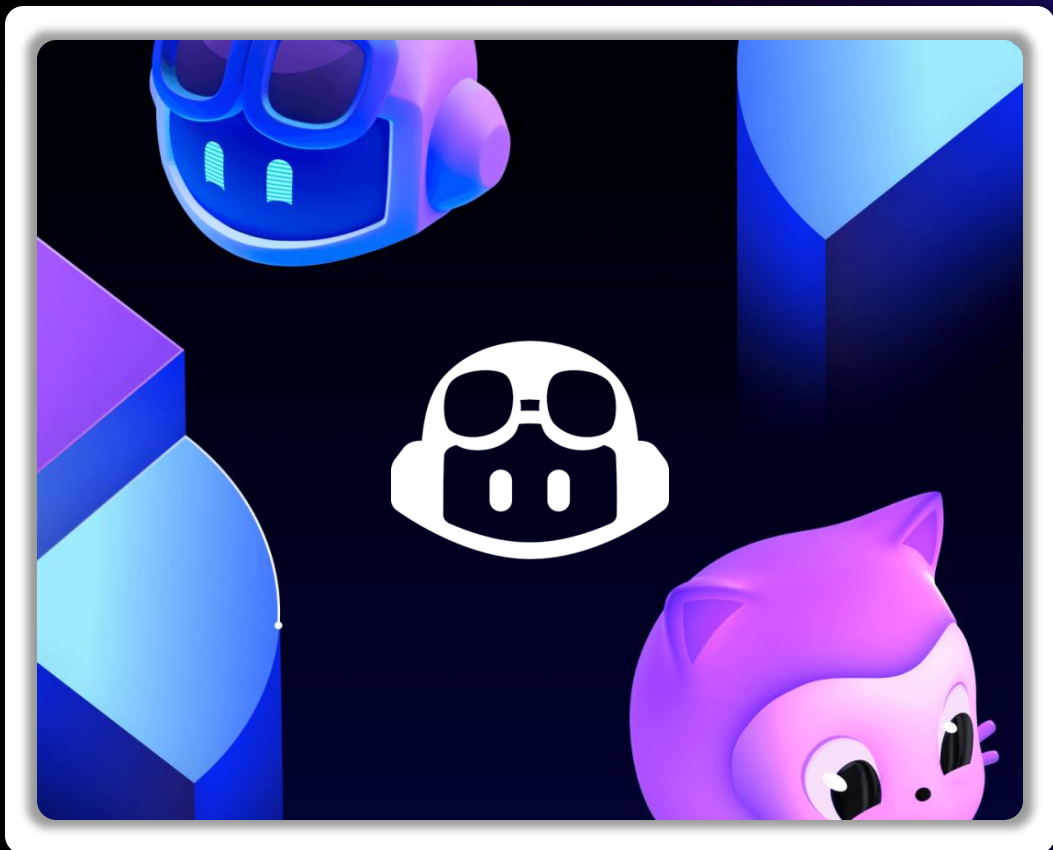


Human-in-the-Loop

- Approve/deny steps and modify the plan
- Provide domain context and constraints
- Override/stop actions; request explanations



What is the GitHub Copilot Coding Agent?



What is the **GitHub Copilot Coding Agent?**

The GitHub Copilot Coding Agent is an autonomous AI developer that works independently in the background to complete larger, multi-step tasks assigned through GitHub issues, chat prompts, or comments.

GitHub Copilot Coding Agent



Lives In The Cloud

Given an issue, will develop, run tests/linters, and open a PR in the background



Repo-Aware

Uses repository context to make changes and presents clean diffs, summaries, and PRs



Create and Execute End-to-End Tests

Explore code, make changes, executes automated tests in an ephemeral development environment

How It Works (At a Glance)

- **Define the Task**

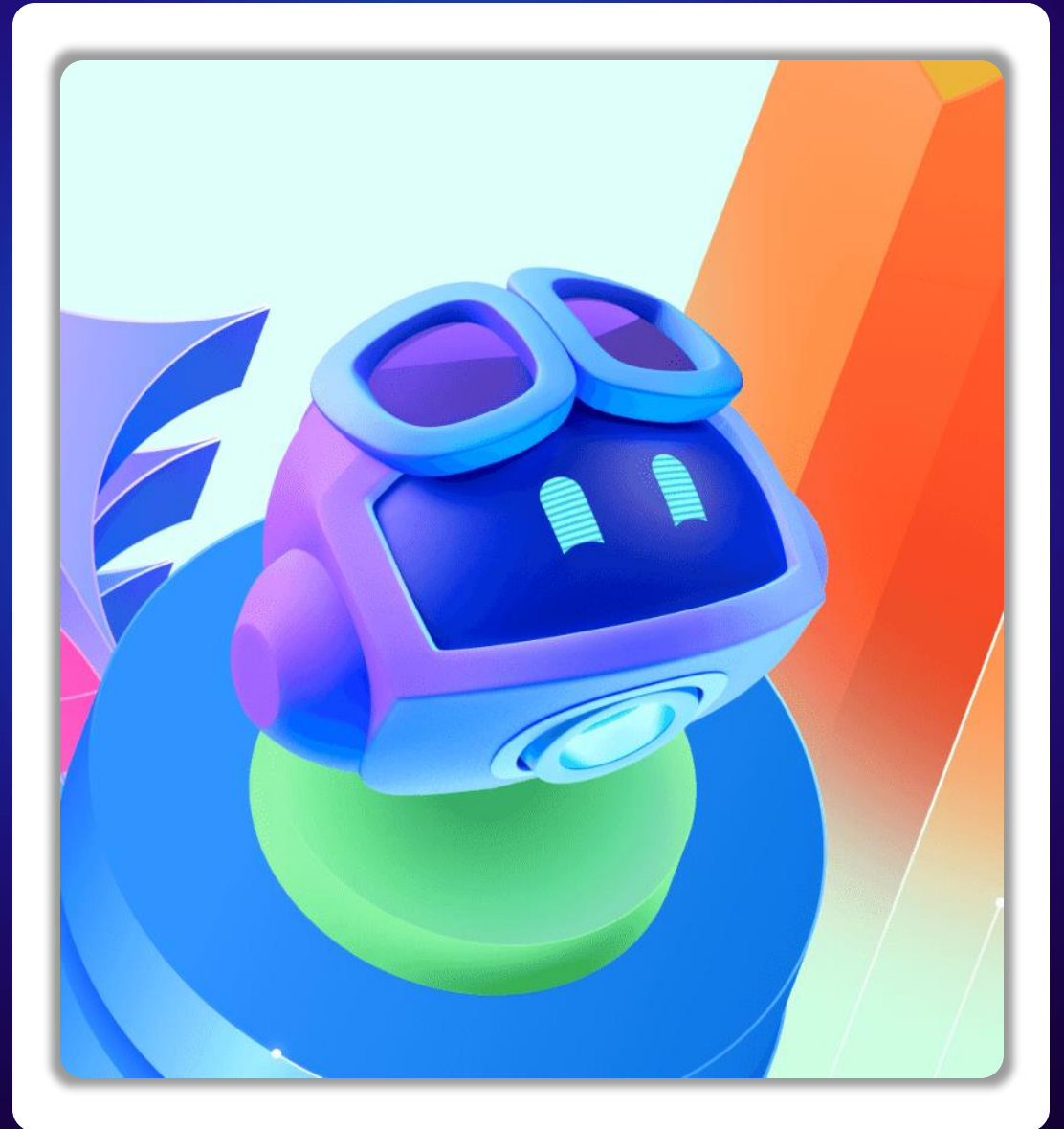
Open an issue or prompt with context, constraints, and a clear Definition of Done

- **Agent Runs in the Cloud**

Plans steps, edits code across files, runs tests/linters/build, retries on failures.

- **You Review and Ship**

Inspect diffs, logs, and the PR summary; request changes or approve and merge.



Allow users to pin side panels #1524

 Open Copilot wants to merge 0 commits into `main` from `feat/pin-side-panels` 



Copilot  commented

This PR modifies the positioning of the pin button in the side panel.

Closes  [Allow users to pin side panels #1520](#)

- ☐ User can pin only one side panel
- ☐ The pinned panel is remembered for the user so it's re-pinned in future visits to the Files Changed page
- ☐ The pinned panel can be resized

 Copilot linked an issue  [Allow user to pin side panels #1520](#)

 Open

 Copilot started work on behalf of [soapoperadiva](#) via [issue assignment](#)

[View session](#)

 Copilot pushed 2 commits

  Introduce pinning state  3fbdc08

 Copilot finished work and requested a review from [soapoperadiva](#)

Assignees



Assign up to 10 people



 Type or choose a user

  **Copilot** Your AI pair programmer



rodbotas6 Rodrigo Botas



chandriverse Chandra Gupta



semyonrush Semyon Maslov



magnusflare Ólaf Magnússon



Prerequisites, **Access Requirements, & Limitations**



GitHub

Subscription Tier

The Agent is a Premium Feature

The Coding Agent is not included with the standard individual Copilot plan.

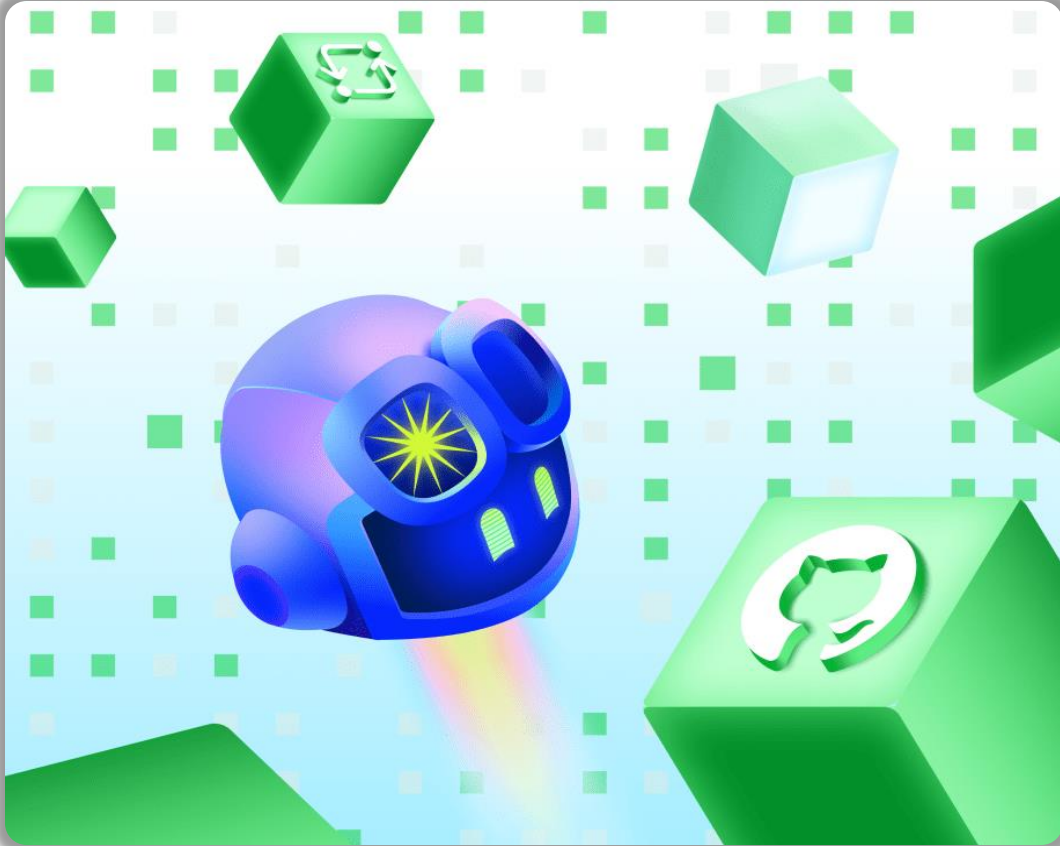
GitHub Copilot Pro

GitHub Copilot Pro+

GitHub Copilot
Business

GitHub Copilot
Enterprise





Cost Considerations

The agent uses **Copilot premium requests** and consumes **GitHub Actions minutes**, which may impact your budgeting.

Each plan includes a **fixed number of premium requests** per user per month. You can purchase additional premium requests for \$0.04 each.

Pro Tip: Set a budget to receive alerts when you reach 75%, 90%, or 100% of your budget.

Coding Agent: Boundaries

Copilot coding agent has certain limitations in its software development workflow and compatibility with other features.



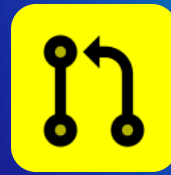
Same Repository

Copilot can only make changes in the same repository where it is creating its pull request.



Limited Context

Copilot can only access context in the same repository as the assigned issue.



One Pull Request

Copilot can only open one pull request at a time.



No Existing PRs

Copilot cannot work on an existing pull request that it did create.

Coding Agent **Limitations**

Important Limitations to Be Aware of When Working with the Coding Agent



Unsigned Commits

Commits aren't signed so signed-commit branches will require history rewrite



Content Exclusions Ignored

Agent doesn't honor Copilot content-exclusion rules.



Cannot Use Self-Hosted Runners

Runs only on GitHub-hosted runners so you can't use your self-hosted runners



GitHub-Only Repos

Works only on repositories hosted on GitHub and not a locally hosted GitHub Enterprise



No Managed-User Personal Repos

GitHub-hosted runners are not available to repositories owned by managed user accounts



No LLM Model Choice

You cannot switch between or pick a model to use – GitHub might change at any time.



Coding Agent vs. Agent Mode

“ ”

“Give a dev a code completion and they’ll merge once.

Teach a dev to wield an AI agent and they’ll empty the backlog before the coffee cools.”

- GitHub



An Asynchronous Teammate that Lives and Develops In The Cloud

Lives within your GitHub repos and can take on issues assigned to it. It will edit code in its own branch, run tests for validation, and open a pull request so you can review before merging.

VS.



A Real-Time Collaborator that Sits Directly in Your Editor

Works inside of VS Code that can provide code suggestions, run tests, and answer questions given a natural-language goal. All changes are made locally.

Pro Tip: You can, and should, use both agents!

Agent Mode



The screenshot displays a code editor window titled "Untitled (Workspace)". The editor shows a Python file named `main.py` with the following code:

```
1 # Entry point for the block game
2
3 import pygame
4 import os
5 from game import Game
6 from ui import draw_board, draw_block, draw_text, draw_line_flash, dr
7 from config import WINDOW_WIDTH, WINDOW_HEIGHT, BLOCK_SIZE, FPS, COLO
8
9
10
11
12 pygame.init()
13 screen = pygame.display.set_mode((WINDOW_WIDTH, WINDOW_HEIGHT))
14 pygame.display.set_caption('RPT Games')
15 clock = pygame.time.Clock()
16
17 # Set window icon after display is initialized
18 icon_path = os.path.join(os.path.dirname(__file__), 'rpt.png')
19 try:
20     # Load rpt.png
21     icon_img = pygame.image.load(icon_path).convert_alpha()
22     # Create orange background
23     ORANGE = (233, 113, 50)
24     icon_size = 128
25     margin = 16
26     logo_max_size = icon_size - 2 * margin
27     # Scale logo to fit within margins
28     logo_w, logo_h = icon_img.get_size()
```

The right sidebar contains a "CHAT" panel with the following text:

Build with agent mode.

AI responses may be inaccurate.

[Generate instructions](#) to onboard AI onto your codebase.

At the bottom of the chat panel, there is a section for adding context and selecting an agent:

Add Context... `main.py` +

Add context (#), extensions (@), commands (/)

Agent `GPT-5 mini`

Agent Mode



▼ block-game ●

- > __pycache__
- config.py
- game.py
- main.py 1, M
- menu.py
- README.md
- requirements.txt
- rpt.png
- ui.py

```
73 fall_event = pygame.USEREVENT + 1
74 pygame.time.set_timer(fall_event, 500)
75
76 running = True
77
78 if logo_loaded:
79     print("Logo loaded successfully.")
80
81 prev_score = game.score
82 while running:
83     for event in pygame.event.get():
84         if event.type == pygame.QUIT:
85             running = False
```

Coding Agent



btkrausen / block-game

Q Type / to search

<> Code

Issues

Pull requests

Actions

Projects

Security

Insights

Settings

Create new issue

Add a title *

Add next block piece for the player

Add a description

Write

Preview

H

B

I

<>

@

When playing the game, it would be great for the next block to be shown to the player so they can play ahead. The block should be shown on the top right with a title of `Next Block: `.

Assignees

Assign up to 10 people to this issue

Q Filter assignees

Group assignees

☒

btkrausen

Suggestions

☐

Copilot

Your AI pair programmer

bot

No milestone

Coding Agent



-  **Copilot** (AI) and others added 3 commits [25 minutes ago](#)
-   Add dark mode toggle with iPhone-style switch ... [20bf2d8](#)
 -   Improve block colors for better contrast in both modes ... [294acc4](#)
 -   Update README with dark mode documentation ... [feaa9d6](#)
-   **Copilot** (AI) changed the title [\[WIP\] Add dark mode feature for game background](#) Add Dark Mode Toggle with iPhone-Style Switch [14 minutes ago](#)
-   **Copilot** (AI) requested a review from **btkrausen** [14 minutes ago](#)
-  **Copilot** finished work on behalf of **btkrausen** [14 minutes ago](#)
-   **btkrausen** marked this pull request as ready for review [now](#)



Review requested

Review has been requested on this pull request. It is not required to merge. [Learn more about requesting a pull request review.](#)

 1 pending review >

Coding Agent vs. Agent Mode

Summary



GitHub Copilot Agent Mode

Lives in your editor for autocompletions, chat, file editing, and more.



GitHub Copilot Coding Agent

Lives in the cloud for working assigned issues, testing, and creating pull requests.



DEMO: Coding Agent

THANK YOU



INSTRUCTOR

**Bryan
Krausen**

